



I'm a student who loves building, solving problems, and creating things that make life easier. My passion for machines started early, and it's only grown stronger as I've learned more.

LINEAR TRANSPORT SYSTEM PROTOTYPE



KEY TECHNICAL FEATURES	ENGINEERING CHALLENGES & SOLUTIONS
• **DUAL MOTOR DRIVE SYSTEM:** Syncronized high-torgue motors for balanced power distribution and reduved belt slippage. • **Reinforced Chassis Design:** A Robust Technic beam structure designed to minimize vibration during high cycle load reversals.	• **CHALLENGE:** Initial belt tension issues causing dynamic instability • **SOLUTION:** Iterated pulley and track configurations to optimize traction and alignment under varying loads. • **CHALLENGE:** Managing high-speed inertial forces. • **SOLUTION:** Implemented structural cross-bracing and a tuned gear-train to absorb kintetic energy

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MECHANICAL LOGIC EXPLORATION



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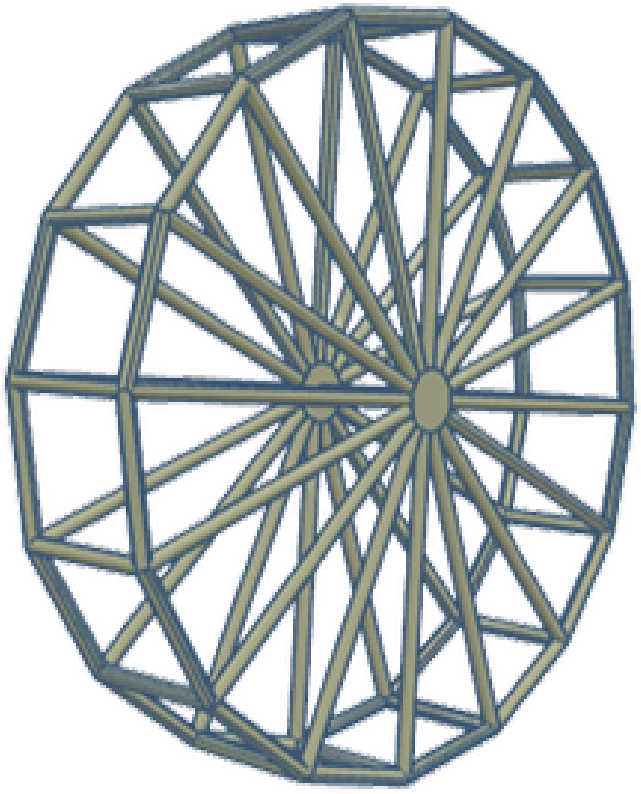
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EARLY ERA

SELF-STABILIZING DYNAMIC ROLLING STRUCTURE







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ENGINEERING BEYOND PHYSICAL CONSTRAINTS

The Shift to Virtual Simulation
While physical builds provided a tactile foundation in gear ratios and structural integrity, the transition to high-fidelity physics engines allowed for a broader exploration of complex systems. By utilizing platforms like Kerbal Space Program, Scrap Mechanic, and modded Minecraft, the focus shifted from "gaming" to iterative digital modeling and systems engineering.

Core Competencies in Virtual Environments

- **Aerodynamics & Orbital Mechanics:** Utilizing Kerbal Space Program to calculate Thrust-to-Weight ratios (TWR), Delta-V requirements, and atmospheric drag coefficients.
- **Complex Logic & Automation**: Leveraging Scrap Mechanic and Minecraft (via Geyser/PaperMC frameworks) to design automated AI-driven workflows and mechanical logic gates.
- **Rapid Iteration & Resource Management**: Using virtual sandboxes to prototype large-scale structures and complex drivetrains where physical material costs would otherwise be prohibitive.

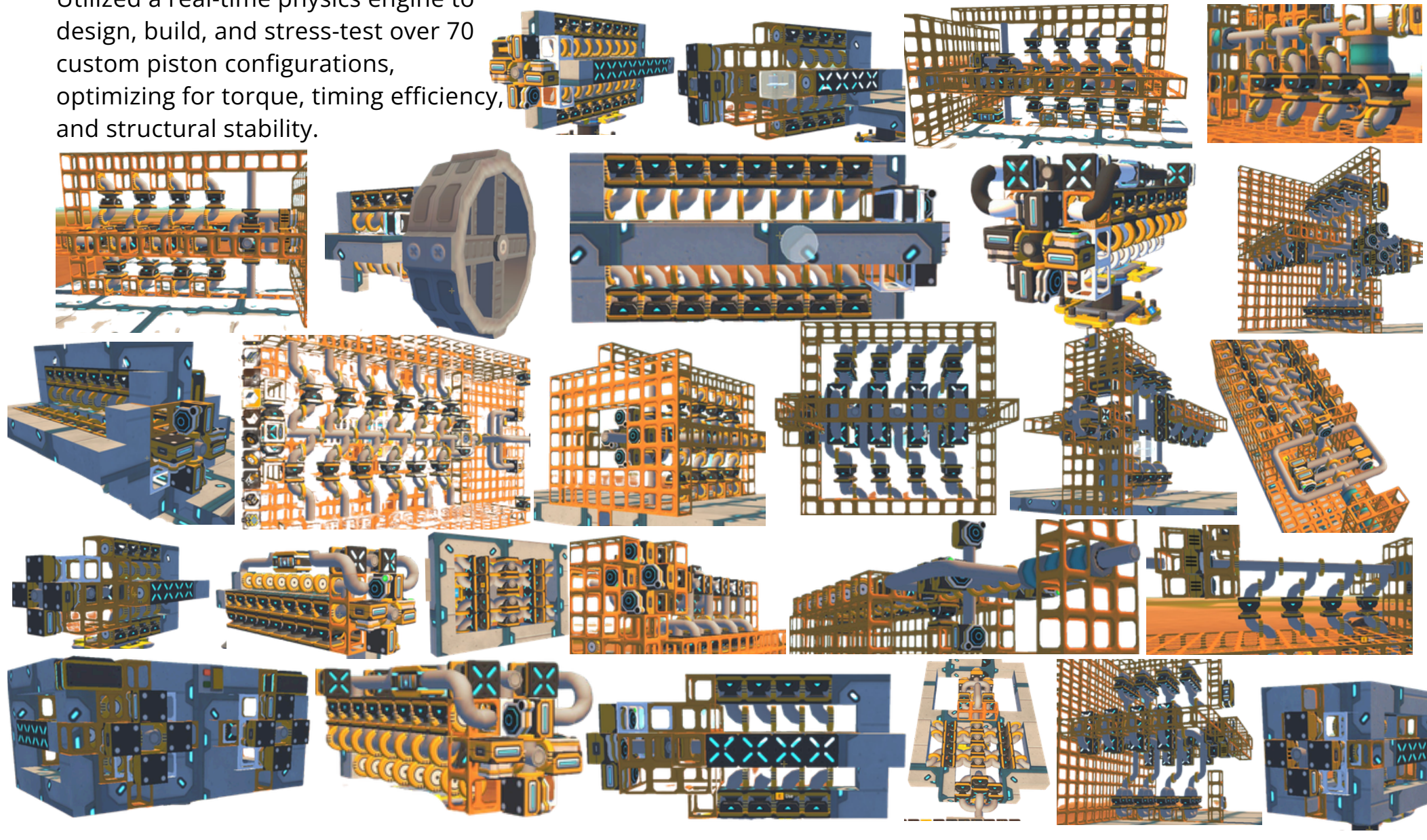
Simulation as a 3D Modeling Tool
Rather than passive entertainment, these platforms served as a dynamic CAD environment. Each "build" required:

- Requirement Analysis: Defining the mission or mechanical goal.
- Simulation Testing: Identifying structural failure points under simulated stress and gravity.
- Optimization: Refining gearboxes and power-to-mass ratios based on real-time physics feedback.

Technical Versatility
This "Virtual Era" directly bridges the gap between physical assembly and professional software development. Aiden’s experience in these engines mirrors the industry-standard use of Digital Twins, where mechanical systems are perfected in a virtual space before ever reaching a physical workbench.

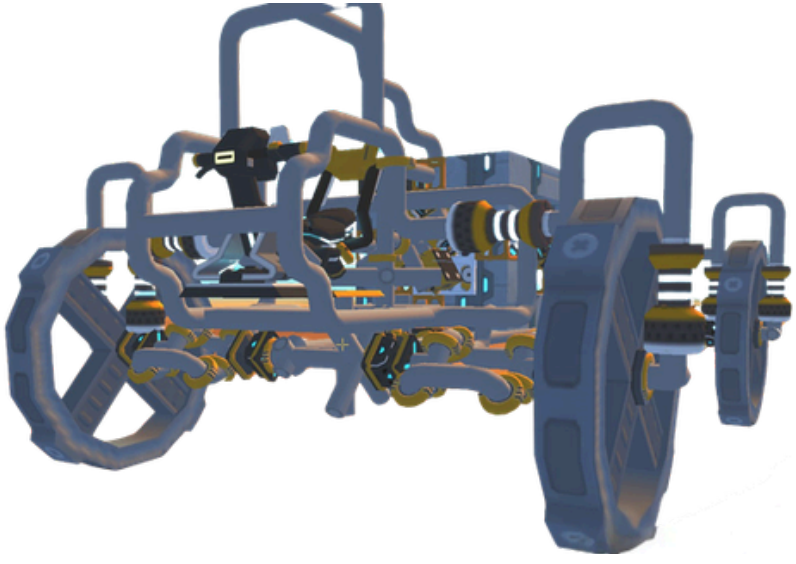
VIRTUAL PROTOTYPING: PISTON ENGINE ITERATION

Utilized a real-time physics engine to design, build, and stress-test over 70 custom piston configurations, optimizing for torque, timing efficiency, and structural stability.

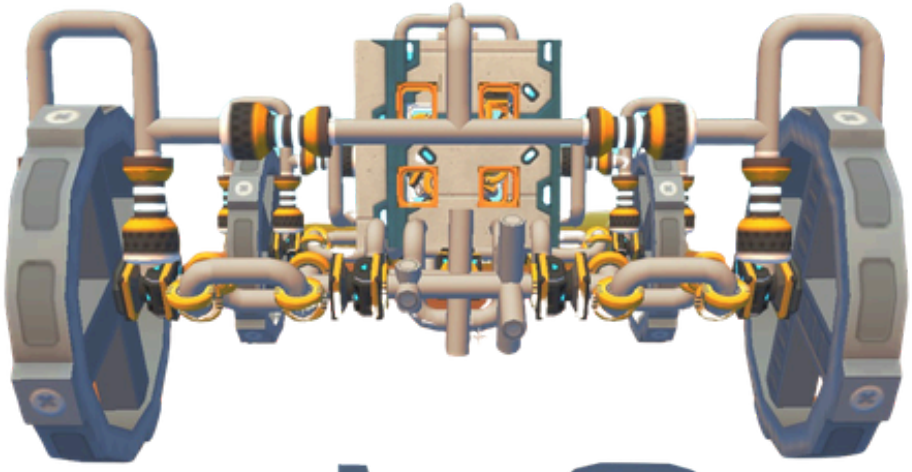


MECHANICAL AUTOMOTIVE DESIGN

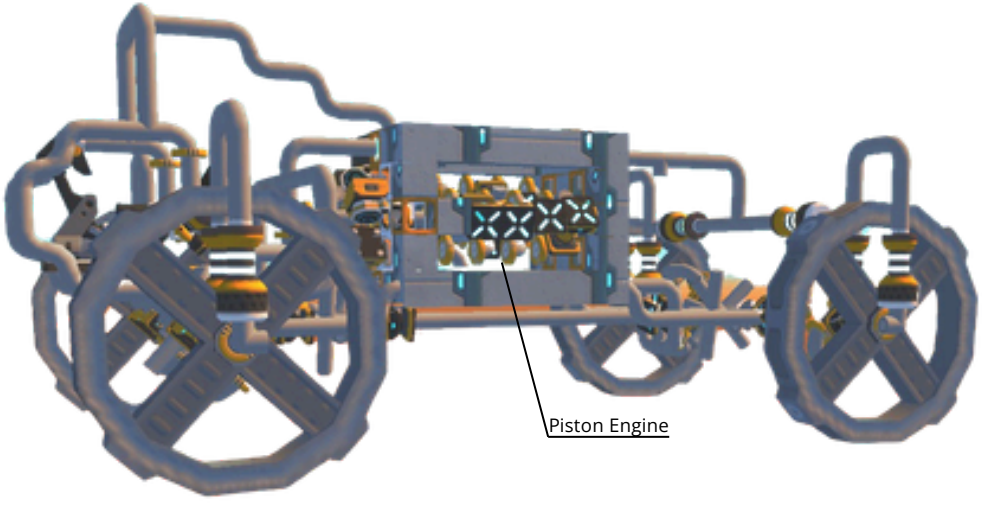
FRONT

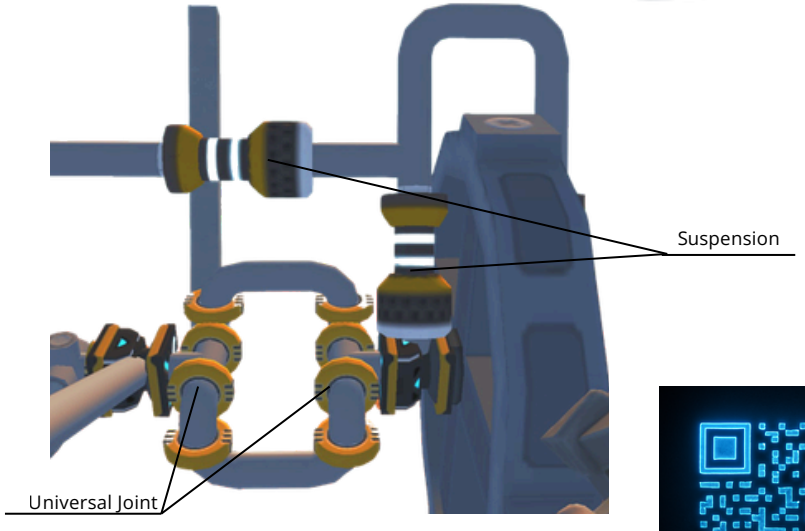


BACK



SIDE



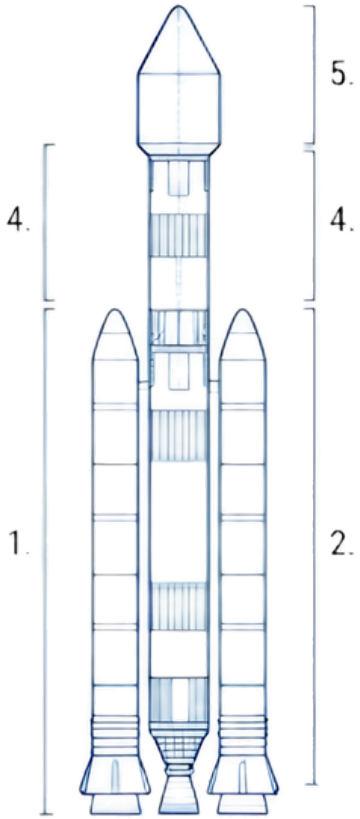


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VIRTUAL ERA

CASE STUDY: SIMULATED MUN MISSION (KSP)



1. Ascent Stage (SRB)
2. Booster Core
3. Trans-Munar Injection (TMI) Stage
4. Munar Descent & Ascent Stage (Lander)
5. Command Module & Recovery

Total Δv:	9,243 m/s
Part Count:	147
Stages:	5



Stage 1-2 seperation

SECTION 1: DESIGN PHILOSOPHY & MULTISTAGE INTEGRATION

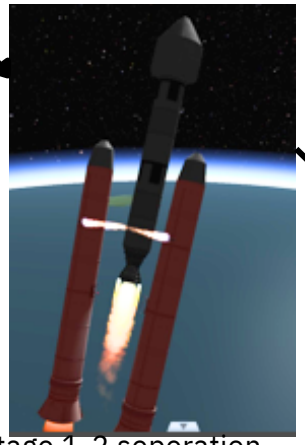
Staging Strategy (Maximizing Vacumn ISP)

- The brilliant orange sun dipped slowly below the jagged horizon, casting long, eerie shadows across the silent valley while the weary travelers finally reached their camp.
- The brilliant orange sun dipped slowly below the jagged horizon, casting long, eerie shadows across the silent valley while the weary travelers finally reached their camp.

Gravity Turn Optimization

The brilliant orange sun dipped slowly below the jagged horizon, casting long, eerie shadows across the silent valley while the weary travelers finally reached theircamp.

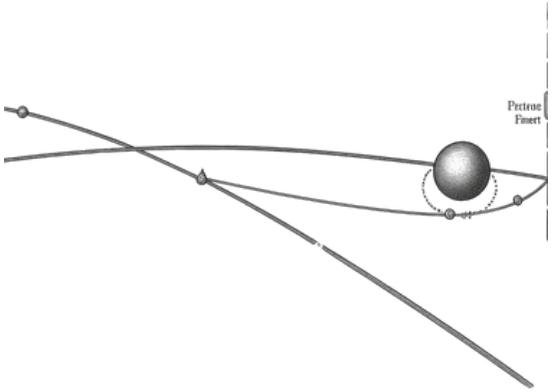
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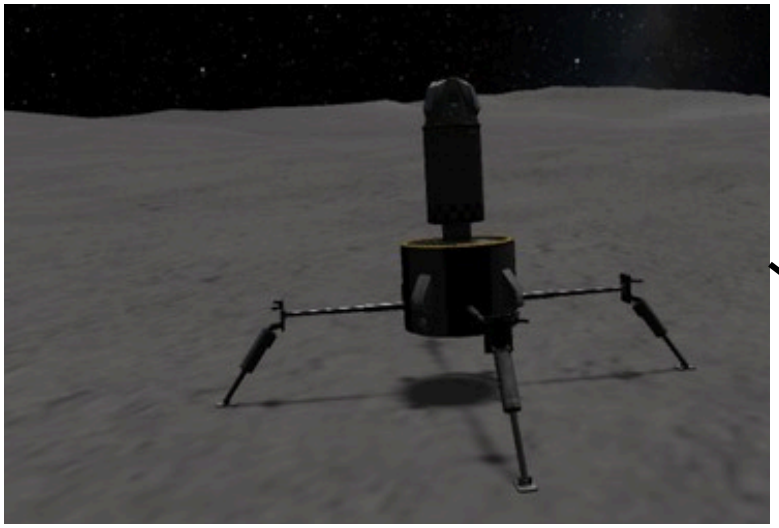
Stage 1-2 seperation



High TWR atmospheric escape

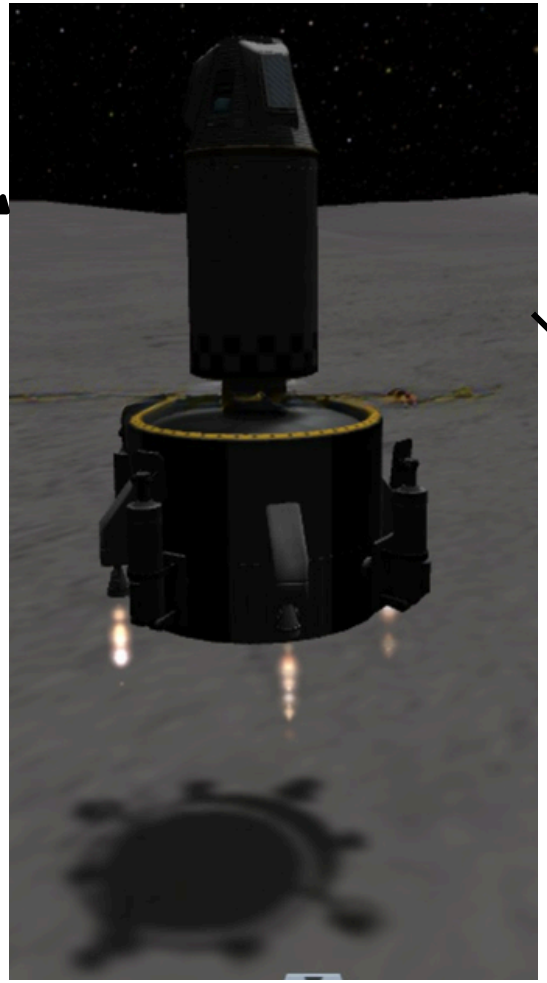


CASE STUDY: SIMULATED MUN MISSION (KSP)



SECTION 2: LANDING DYNAMICS & Re-Entry Systems

- The brilliant orange sun dipped slowly below the jagged horizon, casting long, eerie shadows across the silent valley while the weary travelers finally reached their camp.
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SECTION 3: LANDING DYNAMICS & RE-ENTRY SYSTEMS

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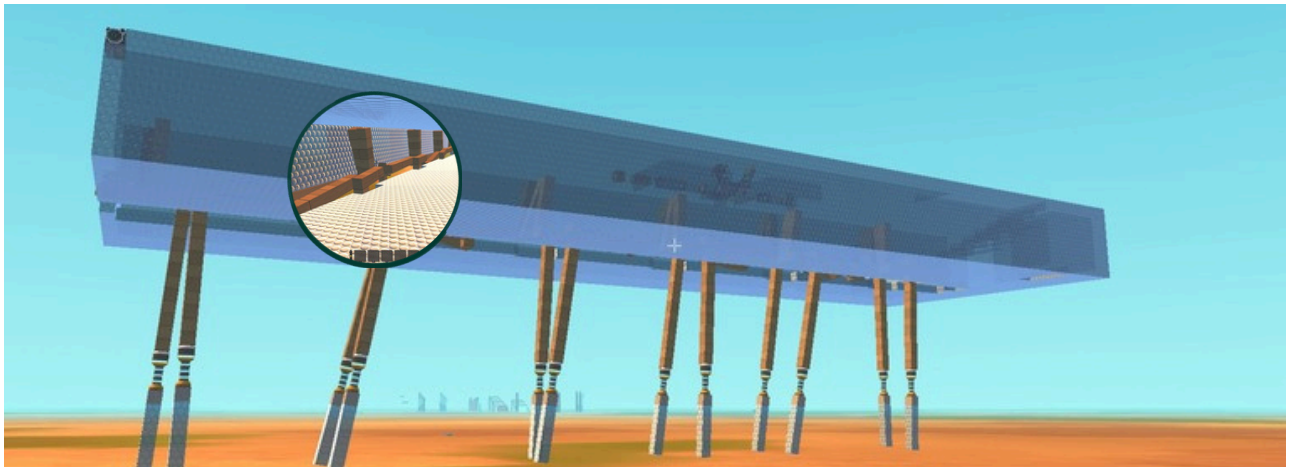
Conclusion & Simulated Systems evaluation

- Real world applications: Staging logic for planetary exploration;

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VIRTUAL ERA

BIOMIMETIC ROBOTICS & LOGIC-DRIVEN KINEMATICS



Designed a multi-legged walking mechanism that integrates sequential rotation matrices and an on-board digital logic array to coordinate joint phase shifts. This mechatronic system synchronizes mechanical linkages to translate rotary motion into a stable linear gait while optimizing runtime timing to prevent joint binding.



Digital Logic Architecture

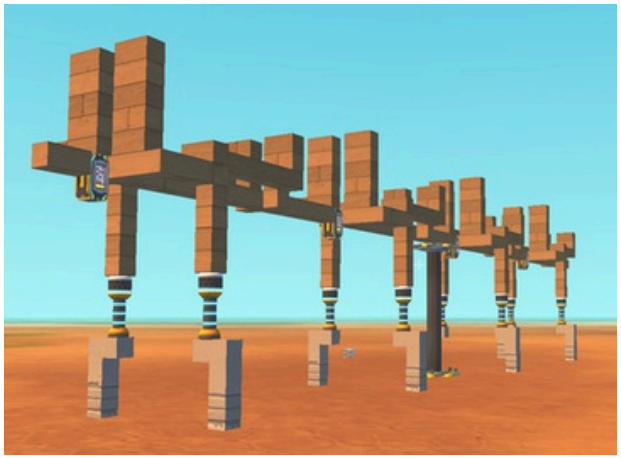
Uses standard Boolean logic gates (AND, OR, XOR, NAND) to handle state machines or control loops for the walking sequence.

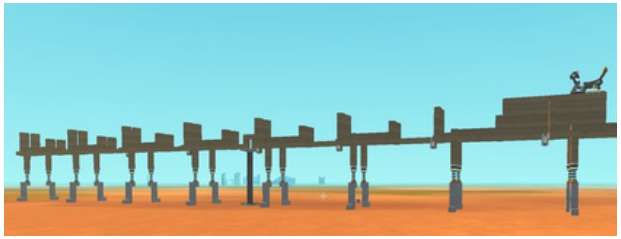


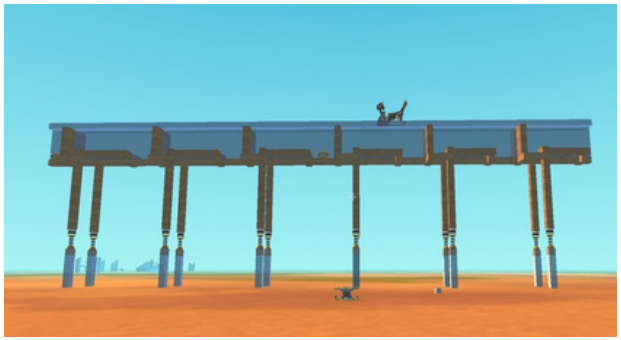
Sequential Multi-Axis Actuation

this functions exactly like a real-world programmable microcontroller (like an Arduino) to time the specific rotational angles of the hip and knee joints

Iterative Prototyping & Gait Optimization







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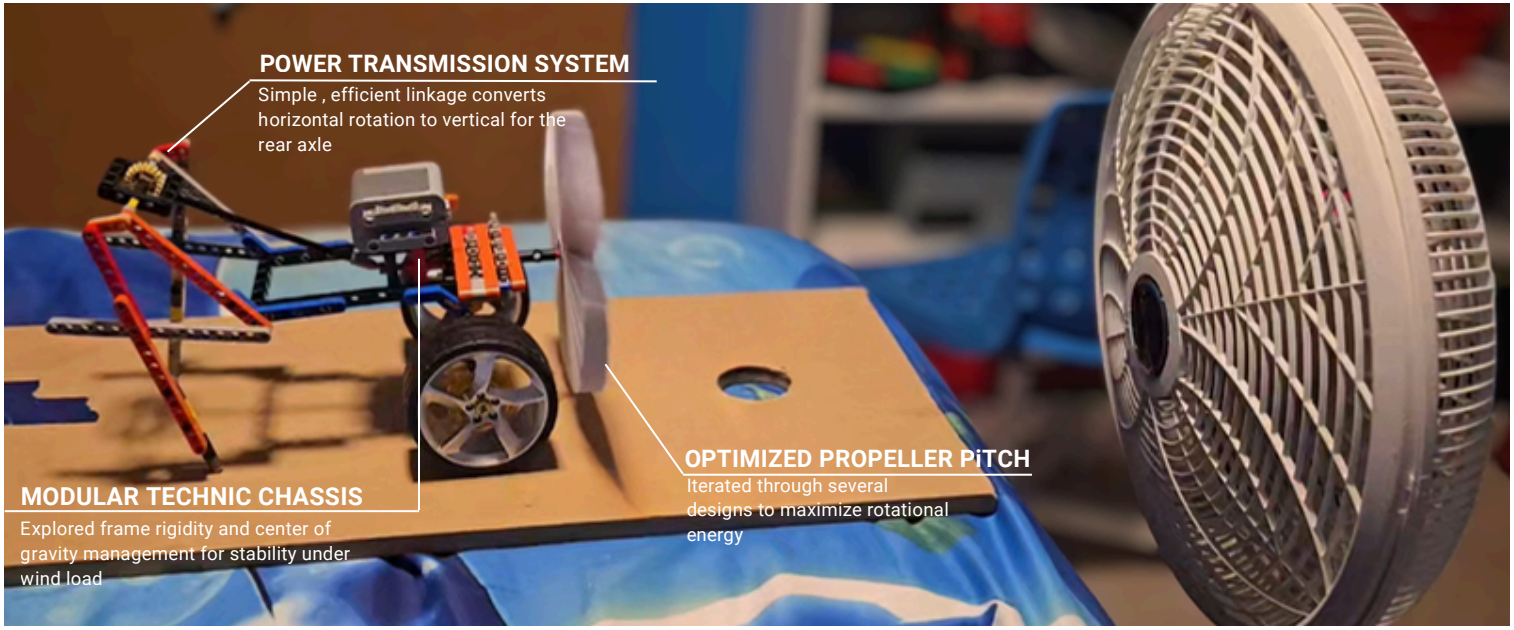
FUSION DESIGN

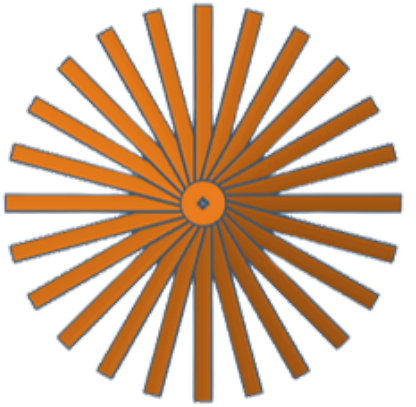




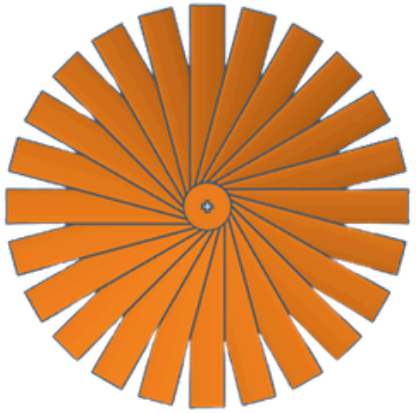
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WIND-POWERED KINEMATIC TRANSPORT PROTOTYPE





The first attempt



The second attempt

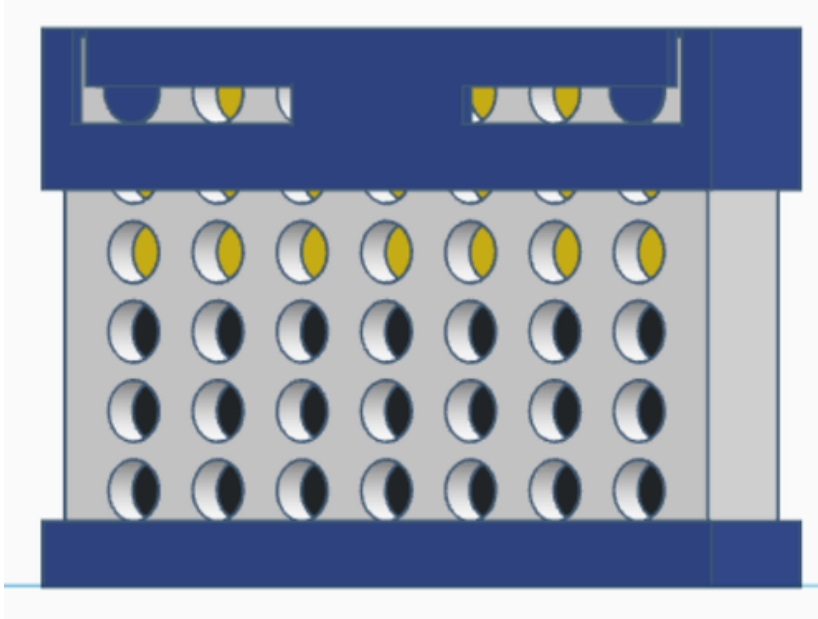
DESIGN INNOVATIONS AND CHALLENGES

- Developed Specific Propeller to harness wind power
- Overcame center-of-gravity imbalance that caused initial tipping
- Implemented a simple but effective mechanical linkage for power transfer
- Utilized Autodesk Tinkercad to iteratively design and propeller blade pitch, followed by manufacturing using a Creality 3D Printer to validate fluid-dynamic theories in a physical environment.

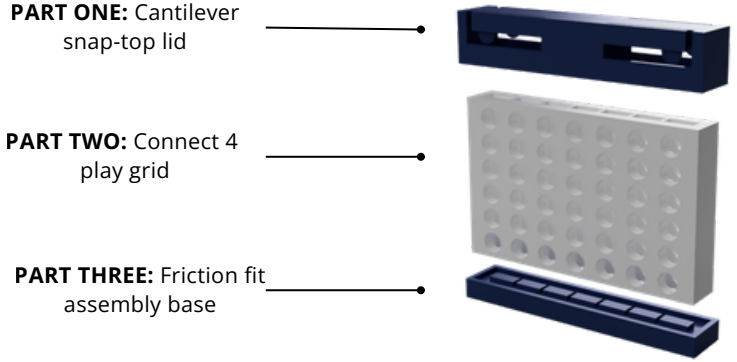
INTEGRATED FASTENING SYSTEMS: THE POCKET CONNECT 4 DESIGN STUDY

The Problem Statement:
Addressing the lack of portability in traditional tabletop games, specifically focusing on component retention during transport.

The Solution: A compact, 3D-printed enclosure featuring an integrated "Snap-Fit" lid mechanism designed for secure travel.



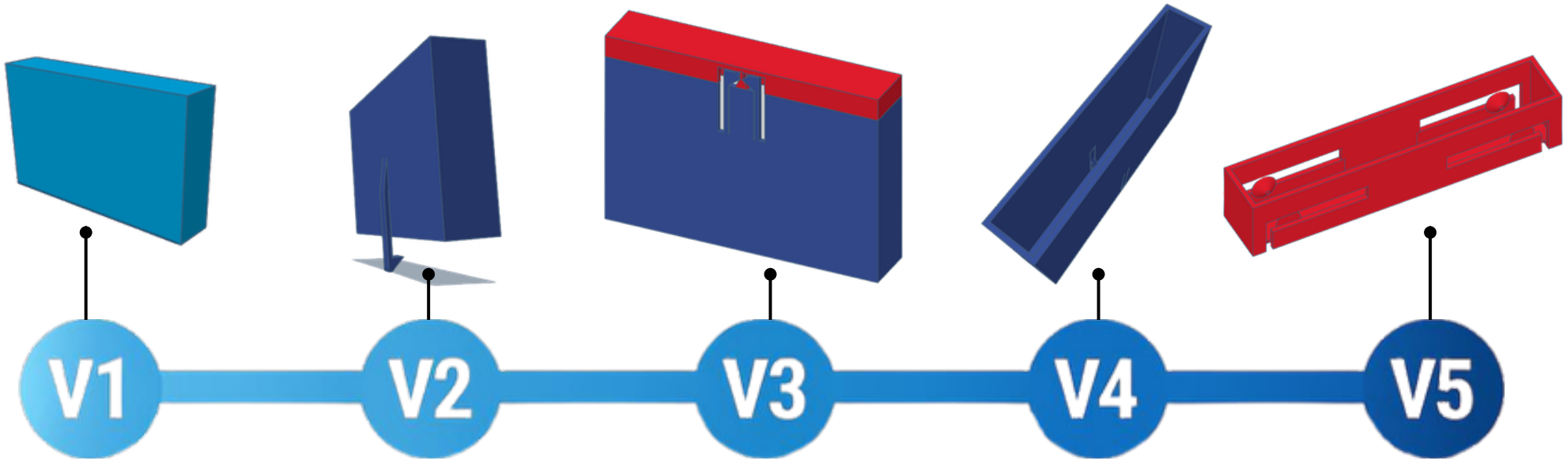
TECHNICAL SPECIFICATIONS	
SOFTWARE	TinkerCAD
HARDWARE	Creality Ender 3 V3 KE
MATERIAL	PLA
TOLERANCE	0.22mm air gap



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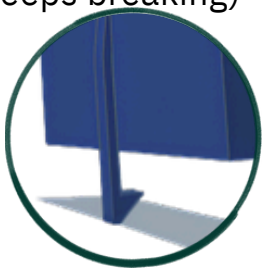
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ITERATIVE PROTOTYPING: EVOLUTION OF THE SNAP-FIT DESIGN.



LOOSE CASE
(Too loose, No snap)

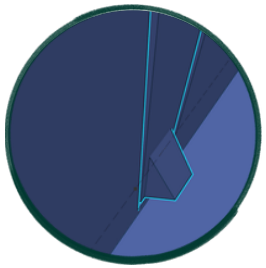
CASE HOOK
(Hook is brittle and keeps breaking)



PINCHER CASE
(Too hard to open)



PINCHER CASE
(Too hard to open)

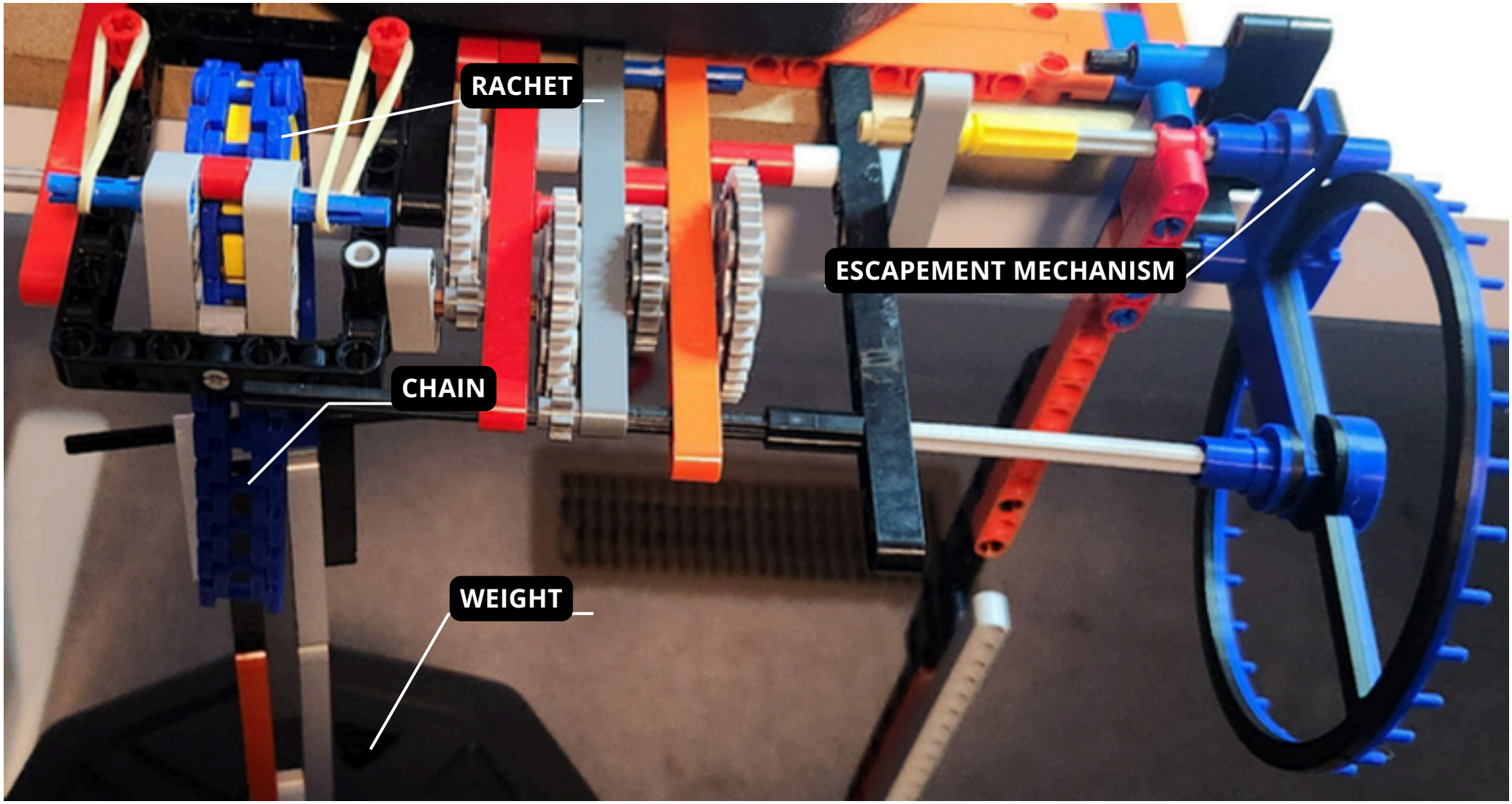


PINCHER CASE
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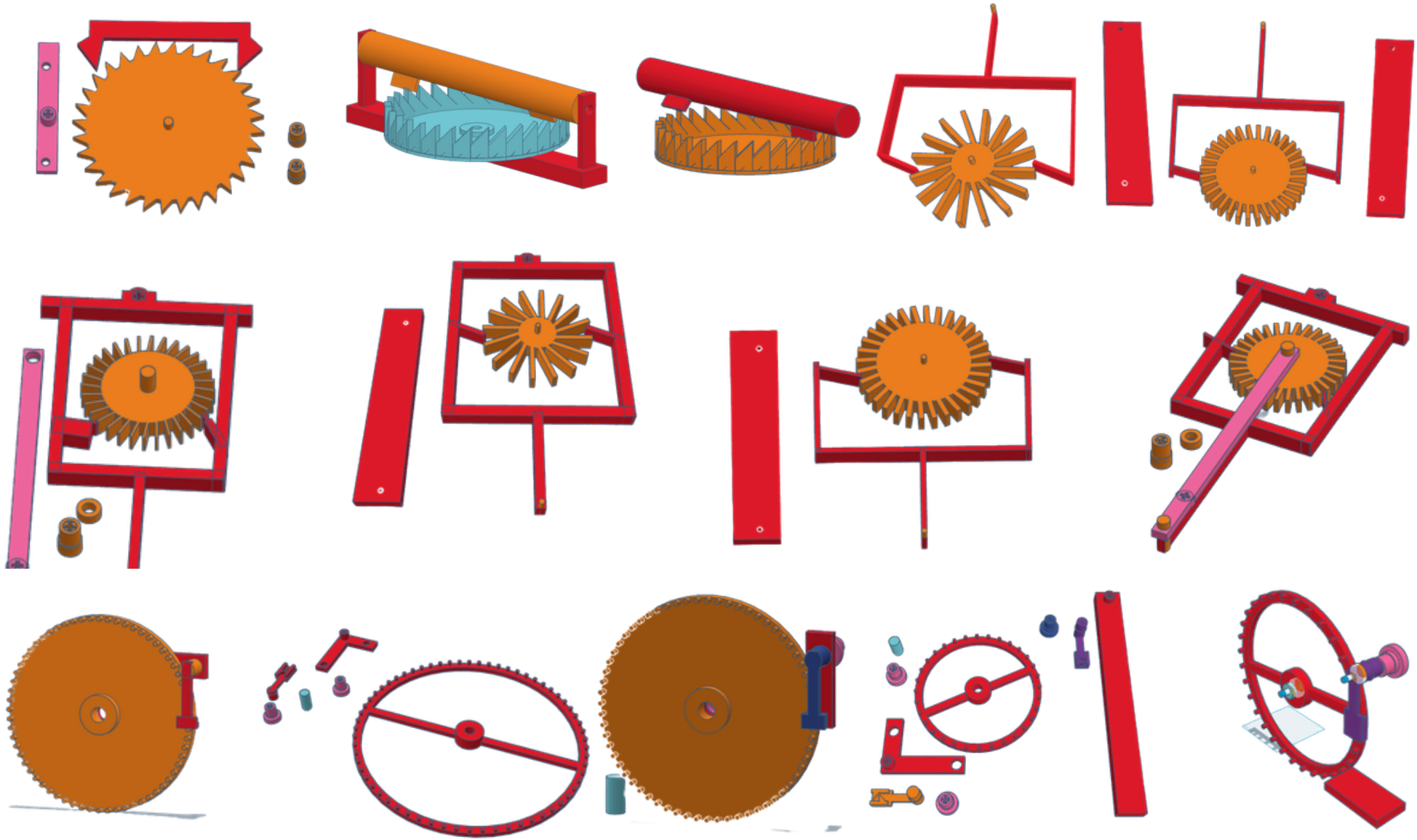


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GRAVITY-DRIVEN ESCAPEMENT & RATCHET SYSTEM

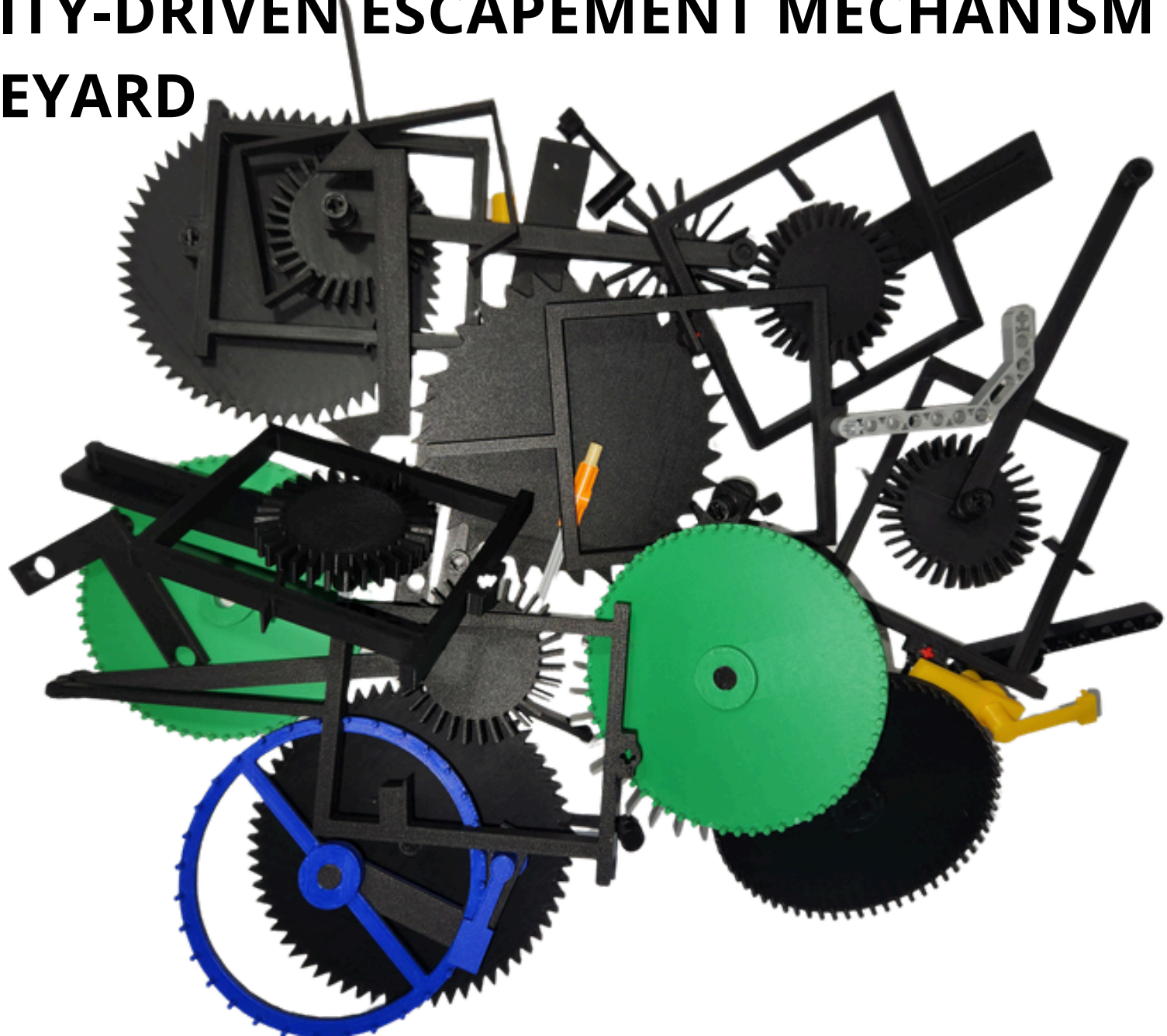


GRAVITY-DRIVEN ESCAPEMENT MECHANISM



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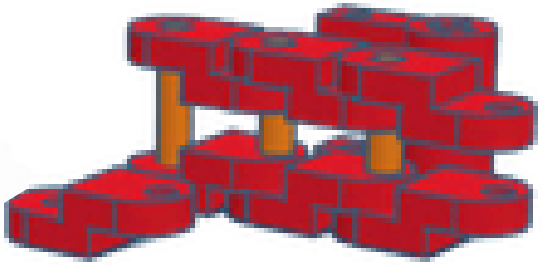
GRAVITY-DRIVEN ESCAPEMENT MECHANISM GRAVEYARD



GRAVITY-DRIVEN PENDULUM RACHET & DRIVE TRAIN



LEGO TREAD



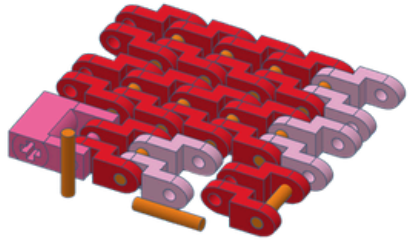
Chain Version 1



Chain Version 2



Chain Version 3



Chain Version 4

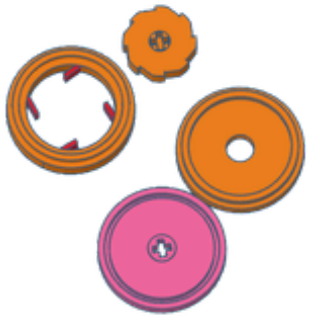




RACHET 1

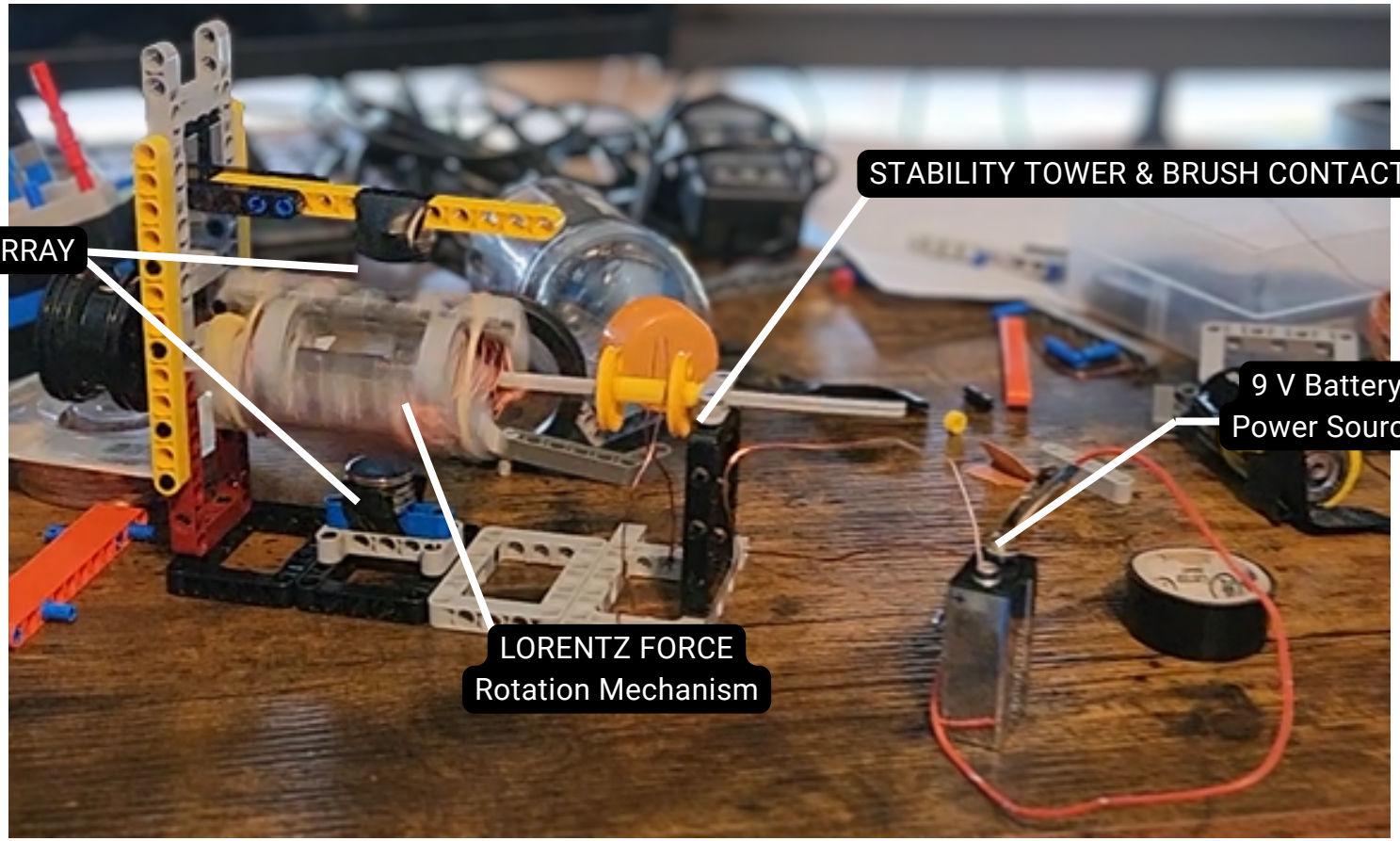


RACHET 2



RACHET 3

BRUSHED DC (BDC) MOTOR PROTOTYPE



PROJECT DESCRIPTION

Designed and assembled a functional brushed DC motor utilizing LEGO Technic for the structural chassis and a hand-wound copper armature. The project required precise alignment of the commutator and brush system to ensure efficient current reversal and consistent torque. This build serves as a proof-of-concept for fundamental electromechanical principles and rotational dynamics.

CHALLENGES AND ITERATIONS

- Swapped initial wiring for high-conductivity magnet wire to improve coil density and electromagnetic flux.
- Calibrated magnetic field strength by testing various neodymium configurations to maximize RPM without stalling.

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Something from Makerspace